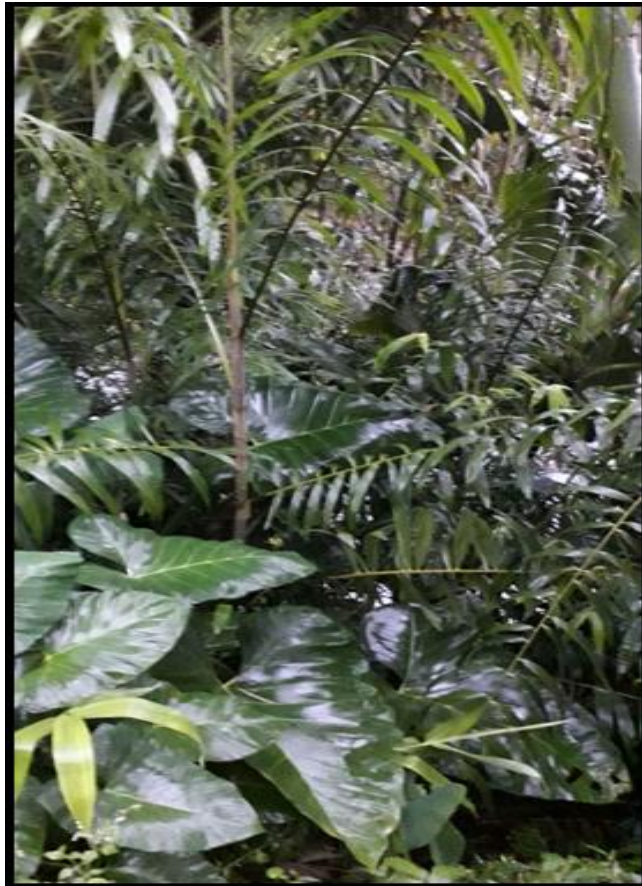


BANA

**Non-Wood Forest Products
Management and Marketing Group**



Lajab Gewog, Dagana Dzongkhag

Plan Period: August 2022-July 2027

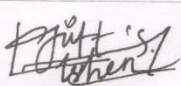
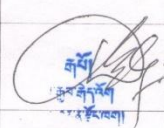
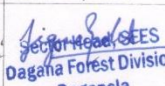
Approval sheet

Contents

APPROVAL SHEET

Name of NWFP Management and Marketing Group:	Bana NWFP Management and Marketing Group
Dzongkhag:	Dagana
Gewog:	Lajab
Village(s):	Salu, Aetshipu
No. of households:	25 nos
Area(Ha)	110.25 Ha
Plan Period:	2021-2026

Prepared and submitted for approval

	Chairperson NWFP Management and Marketing Group	Gup of the Gewog	Head Social Forestry Section of Division/Park	Chief Forestry Officer of Division/Park
Signature:				
Name:			Tigme Wangchuk Kende Dagana Forest Division Dagapela	Chief Forestry Officer Dagana Forest Division Dagapela
Date:				

REVIEWED AND RECOMMENDED FOR APPROVAL

	Reviewing Officer, SFED	Chief Forest Officer of the SFED	Head of the Department
Signature:			
Name:			
Date:			

APPROVED BY

SECRETARY
Ministry of Agriculture and Forests

Approval sheet	i
Part A: NWFP Management and Marketing Plan.....	4
1. Background.....	4
1.1. Goals and Objectives	4
1.2. Description of the NWFPs.....	5
2. Harvesting prescription for sustainable harvesting of <i>Plectocomia himalayana</i>	6
3. The NWFP Collection Area.....	7
4. Marketing and Sales Plan.....	7
5. Marketing arrangement.....	8
6. Benefits to members	8
7. Cane Resource assessment.....	9
8. Analysis of Resource Assessment Data	9
9. Annual Harvesting Limits.....	9
a. Cane population and annual harvesting limit of block I.....	10
b. Annual Harvesting Limits of Canes from block II	11
c. Annual Harvesting Limits of Canes from block III.....	12
PART B: By-Laws: Group Regulations and Norms.....	13
10. Members of the NWFP management & marketing group	13
10.1 Responsibilities and rights of members.....	13
11. Executive committee.....	13
11.1. Chairperson	13
11.1.1 Responsibilities of Chairperson	13
11.2. Secretary	14
11.2.1 Responsibilities of Secretary.....	14
11.3. Treasurer	14
11.3.1 Responsibilities of Treasurer	14
12. Compensation for executive Committee members	15
13. Membership Fee and purpose	15
14. Community development fund and utilization of fund	15
15. Conflict Resolution	15
16. Penalties	16
17. Authority of the Department.....	16

Annexure I: Membership list	17
Annexure II: Block description form for canes	19
Annexure III: Data Compilation Sheet	22
Annexure IV: Application for the establishment of NWFP group from Gewog Administration ..	29

Part A: NWFP Management and Marketing Plan

1. Background

The Bana Chiwog is located at the heart of Lajab Gewog under Dagana Dzongkhag. The Chiwog has 52 household of Salo and Aetshipu with ethnicity of *Lhotshompas*, *Sharshops* and *Ngalops*. The people in the village grow vegetables, maize, paddy to meet annual household food requirement and cardamom as cash crop. With support from agriculture sector, they also practice modern farming - use of greenhouse and also installed electric fencing to protect their crops from wild animal. They also rear cattle such as cow, oxen and goat, which serve as source of the dairy product, meat and oxen to plough field despite having farm machinery such as power-tiller.

The Gewog, being located on the fringes of the Cool-broadleaved forests, which has a rich biodiversity of the flora and fauna. It favours the growth of high economic valued trees such as walnut and kimbu. Also various animals such as barking deer and Himalayan black bear are also found in the forest. Besides that, non-wood forest products (NWFP) such as *Elatosteme lineolatum*, *Zanthoxylum armatum*, *Persia fruitifera*, *Diplazium esculentum*, bamboos (*Chimonobambusa callosa*, *Dendrocalamus hamiltonii* and *Himalayanacalamus hookerinus*) and *Plectocomia himalayana* are found and being collected by community for self-consumption and also for sale. They depend on cane (*Plectocomia himalayana*) mainly for baskets making and other useful household utilitarian products. Cane shoots are mainly used as food during the gathering and other important events such as marriage and other related celebrations.

Currently the cane resources from the area are collected based on permit system. However, it is not managed properly as there are over exploitation as an ownership is still hold by the government. The formation of the management group would enable handing over of management responsibilities to the community and reduce overharvesting and over exploitation of the resources. Such initiative is likely to increase income generation and employment opportunities for the end users. With control gained over the area, the group can also planned for increasing cane stock through plantation in the feasible area. Based on cane assessment result, the group can harvest 2326 numbers of matured canes 1238 numbers of shoots in a year. The resources growing areas were located after the prior consultation conducted with group.

1.1. Goals and Objectives

1. Sustainable harvesting of matured cane and cane shoots.
2. To develop marketing chain for cane resources. And products
3. Product diversification through skill development program to members.

1.2. Description of the NWFPs

The region harbors growth of variety of Non-Wood Forest Products such as *Elatostema lineolatum*, mushrooms, *Zanthoxylum armatum*, various bamboos, wild avocado, various fern species, and cane; most abundantly *Plectocomia himalayana*, which is also known as the *Patsa*(Dz), Phasey (Sharshop) and Phakeri beet (Lh) that has the distribution range from 1200 m above sea level to 2150 m above the sea level.

During the peak harvesting season, the NWFP of the area are harvested for local consumption as well as for commercial purpose which are sold in Thimphu market by the local vendor directly. Some species of Bamboos such as *Chimnobambusa callosa*, *Dendrocalamus hamiltonii* and *Himalayacalamus hookerinus* are harvested for fencing, and for making various households products which has daily usage.

The shoot portion of the cane species are used as the source of food during some of the important occasions such as marriage party, Londa and other occasional gatherings. However, the matured cane parts are used for the production of baskets and various kitchen utensils.

Table01.Brief description of NWFPs for commercial harvesting

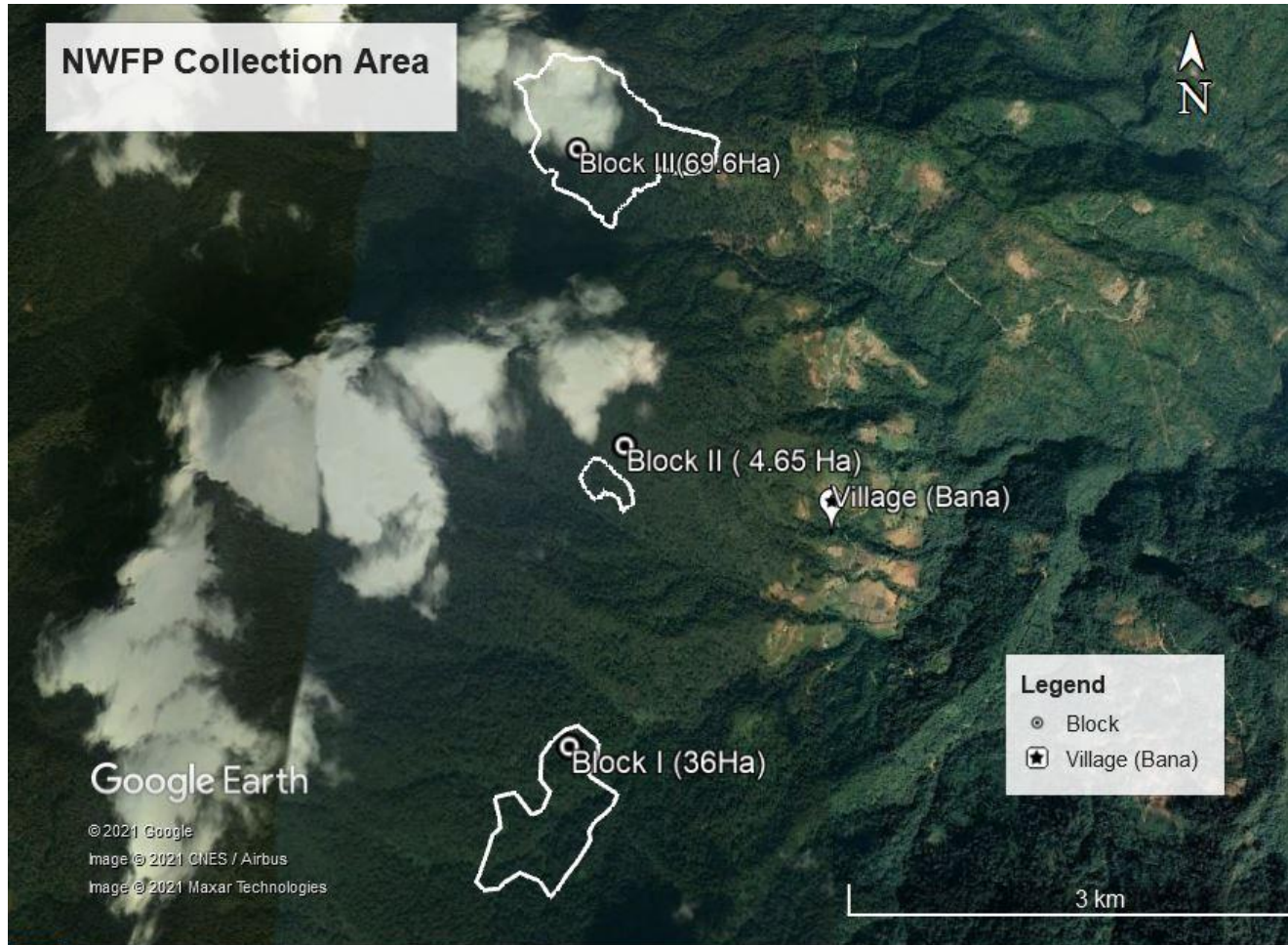
NWFP Name	Harvesting Area	Parts harvested	Uses or Purposes(traditional& others)	Harvesting season (month)	Estimated quantity that can be collected
<i>Plectocomia himalayana</i> (Canes)	Pakhaychok, Sukaypokri, Aetshilum, Dorona and Lalichok	Young Shoots & Matured Canes	Shoots used as cuisine (<i>patsajaju</i>) Matured canes for the handicrafts	Winter for the Handicraft purpose Summer for cane shoots	2820Nos 1630Nos

2. Harvesting prescription for sustainable harvesting of *Plectocomia himalayana*

Scientific Name	Prescription for sustainable harvesting of cane and shoot
<i>Plectocomia himalayana</i>	Collection of matured cane: Harvest during winter as canes have a better quality during this season Pull down as much of the stem as possible (avoid leaving cane parts high up in the tree) Cut only canes with a usable length of at least 10m. Trees supporting the canes shall not be felled
	For cane shoots: Cut young shoots by bending the cane stem with the help of a long and strong stick with a hook. Do not cut the entire cane to collect the shoots. Cut the leading shoot with a sharp knife diagonally (not straight in order to avoid rotting of the remaining stump); leave lateral shoots. Care shall be taken to avoid damaging sucker buds (these will later develop into lateral shoots). Harvest only shoots identified for collection during plan (year 1 to 4 years)

3. The NWFP Collection Area

Below is the map of the NWFP collection area of the Bana Chiwog, which is divided into three blocks. The farthest collection area is located at three hours distance from the village.



4. Marketing and Sales Plan

The cane products will be sold in two forms; 1) Shoots and handicraft. The potential markets for the sale of the canes shoots and its products are Thimphu, Wangdue, Dagana and Tsirang. Shoots will be collected from open patches where cane hasn't climbed on trees. Also it will be collected from cut canes. The handicraft will be produced from matured cane. It will be produced and sold from handicraft house constructed at the Gewog center recently.

The collection and sell of products will take place only after convent of general meeting. The group shall discuss each year's collection timing of cane shoot and matured cane and also finalize group's middlemen. The market for the product will be sought by middlemen identified from the group. He will arrange the transportation and deposit royalty amount

paid to the forest office and other agreed amount to the group account before lifting the products. The quantity of the sales to the specific location shall be as per the demand and if incase lots of demand comes, and then it shall be divided among them. Quantity of cane shoot and matured cane to be collected in plan period and income estimated is given in table 02.

Table 2. Marketing and Sales Plan

NWFP	Potential Market	Estimated quantity for sale per season (unit)	Expected price per unit	Total (Nu.)	Remarks
Cane shoots (<i>Plectocomia himalayana</i>)	In-country, mostly Thimphu and Wangdue	2820	Nu.30/unit to members	84600.00	Deposit 5% of the sales proceeds to group account.
Canes (<i>Plectocomia himalayana</i>)	In-country, mostly Thimphu and Wangdue	1630	Nu.100/unit	163000/-	Deposit 5% of the sales proceeds to group account.
Total Amount				Nu.247,600/-	

Note: The Price of the product shall be fixed from time to time as per meeting when selling to outsiders after receiving an order.

5. Marketing arrangement

The main market shall be Thimphu, Wangdue, Tsirang and Dagana. The group will collect NWFPs resources on order basis. Based on product demand, the chairperson will obtain collection permit from the forest office.

6. Benefits to members

- The members registered in the group shall have the equal right for the collection and marketing of the NWFP as per the guidelines of the Management group.
- Each member has right to get equal share from the accumulated group fund.
- The resources of cane shall be equally divided among the group.

7. Cane Resource assessment

The cane resource assessment was done with the help of the QGIS. The sampling intensity used was 5% or more. The area was first demarcated after the consultation with the user group with the help of the GPS device in the real field, which was later converted into shape file in the QGIS.

The shape file was then inserted in the Q-GIS and the plotting was done. The regular plots were plotted using sampling intensity of 5% or more. The plots were then again inserted into the GPS device for the resource assessment.

8. Analysis of Resource Assessment Data

The data collected in the field was analyzed using the excel sheet where the data of the canes in the different sizes are entered into the different column. After entering the data, area of the total sample plot is drawn. Then the total canes or shoots of the different class or group was then drawn, which was further divided into each sample plots where the data are extrapolated and then the annual harvesting limit of each sample class was drawn.

9. Annual Harvesting Limits

Mature cane category shall be collected based on the harvesting limit set in the management and marketing plan from three blocks. Here the matured cane means cane stem harvestable during survey period and those cane stem which will get matured and cane be harvested during the plan period (5years). An annual harvesting limit of cane in a block has been calculated assuming that it would take 10 years to get usable cane length of 10meters. However, during harvest it shall be taken into consideration that semi-mature canes will only be harvested only towards end of the plan period. Semi mature category of the canes growing on open areas shall be harvested for the shoots during plan period. Under any circumstances, immature canes stems climbing on trees will not be harvested for shoot. The suckers and the cane category of the semi-mature and immature shall be left for the growing that shall replace the one that are harvested. For shoots five years was estimated as its rotation period. In this plan an annual harvesting limit was not calculated sucker as we cannot determine survival of suckers growing in the block.

a. Cane population and annual harvesting limit of block I

Block Area in acres/Ha = 36 Ha/ 88.96 Acres Total sample area covered in block(acres) = (no of sample plots in block x sample plot size)/10,000 = (34 plots x 500.34 m ²)/10,000 = (1.701156 x 2.471) = 4.20 Acres No. of sample plots to be accommodated in block =Block area/total sample area(88.96/4.20=21.18 (c)				
Time period for harvesting canes for handicrafts/cane shoots	Cane species	Cane count in all sample plot (a)	Total cane stock in (c) = (a) x (b)	AHL of from the block Cane=(c/10) Shoot=(c/5)
Mature Canes Year 1-2 (L>20m (Cane > 10m)	<i>Plectocomia himalayana</i>	64	1355.52	136
Semi-Mature Year 3-4 (>20m (cane<10m) Length (Canes)	<i>Plectocomia himalayana</i>	41	868.38	87
Semi-Mature Year 3-4 (>20m (cane<10m) Length (Shoots)	<i>Plectocomia himalayana</i>	48	1016.64	203
Immature Year 5-6 (10-20m) (Canes)	<i>Plectocomia himalayana</i>	21	444.78	41
Immature Year 5-6 (10-20m) (Shoots)	<i>Plectocomia himalayana</i>	30	635.40	127
Immature Year 7-8(2-<10m Length) (canes)	<i>Plectocomia himalayana</i>	23	487.14	49
Immature Year 7-8(2-<10m Length) (Shoots)	<i>Plectocomia himalayana</i>	50	1059	212
Suckers Year 9-10 (Present<2m)	<i>Plectocomia himalayana</i>	148	3134.64	Nil

b. Annual Harvesting Limits of Canes from block II

Block Area in acres/Ha = 4.65 Ha/ 11.5 Acres Total sample area covered in block(acres) = (no of sample plots in block x sample plot size)/10,000 = (5 plots x 500.34 m ²)/10,000 = (0.25017 x 2.471) = 0.61817 Acres (b) No. of sample plots to be accommodated in block =Block area/total sample area(11.5/0.61817=19 (c)				
Population structure in Cane in a block	Cane species	No. of canes in sampled area (d)	Total cane stems in block (e)= d x c	AHL from the block Cane=e/10 Shoot=e/5
Mature Canes Year 1-2 (L>20m (Cane > 10m)	<i>Plectocomia himalayana</i>	59	1121	112
Semi-Mature Year 3-4 (L>20m (cane<10m) Length (Canes)	<i>Plectocomia himalayana</i>	32	608	61
Semi-Mature Year 3-4 (L>20m (cane<10m) Length (Shoots)	<i>Plectocomia himalayana</i>	33	627	125
Immature Year 5-6 (10-20m) (Canes)	<i>Plectocomia himalayana</i>	10	190	19
Immature Year 5-6 (10-20m) (Shoots)	<i>Plectocomia himalayana</i>	9	171	34
Immature Year 7-8(2-<10m Length) (canes)	<i>Plectocomia himalayana</i>	23	437	44
Immature Year 7-8(2-<10m Length) (Shoots)	<i>Plectocomia himalayana</i>	27	513	102
Suckers Year 9-10 (Present<2m)	<i>Plectocomia himalayana</i>	113	2147	Nil

c. Annual Harvesting Limits of Canes from block III

Block Area in acres/Ha = 69.6 Ha/ 172.73 Acres (a) Total sample area covered in block (acres) = (no of sample plots in block x sample plot size)/10,000 = (69 plots x 500.34 m ²)/10,000 = (3.452346 x 2.471) = 8.530749 Acres (b) No. of sample plots to be accommodated in block =Block area/total sample area(172.73/8.530749=20.25 (c)				
Time period for harvesting canes for handicrafts/cane shoots	Cane species	No. of canes in sampling area (d)	Total cane in block (e)= (c x d)	AHL from the block Cane = e/ 10 years Shoot= e/ 5 years
Mature Canes Year 1-2 (L>20m (Cane > 10m)	<i>Plectocomia himalayana</i>	583	11806	1181
Semi-Mature Year 3-4 (>20m (cane<10m) Length (Canes)	<i>Plectocomia himalayana</i>	261	5285	53
Semi-Mature Year 3-4 (>20m (cane<10m) Length (Shoots)	<i>Plectocomiahimalayana</i>	213	4313	862
Immature Year 5-6 (10-20m) (Canes)	<i>Plectocomia himalayana</i>	86	1742	17
Immature Year 5-6 (10-20m) (Shoots)	<i>Plectocomiahimalayana</i>	278	5629	1125
Immature Year 7-8(2-<10m Length) (canes)	<i>Plectocomia himalayana</i>	38	770	8
Immature Year 7-8(2-<10m Length) (Shoots)	<i>Plectocomia himalayana</i>	295	5974	1194
Suckers Year 9-10 (Present<2m)	<i>Plectocomia himalayana</i>	148	2997	Nil

PART B: By-Laws: Group Regulations and Norms

	Village	Number of member households	Gewog
	Salo, Aetshiphu	25	Lajab
10. Members of the NWFP management & marketing group	10.1 Responsibilities and rights of members Every household with definite Gung and Thram from Bana Chiwog are eligible to become a member of NWFP Management Group. Each member will contribute a sum of Nu.100 (one hundred) only as annual membership fee in the month of January every year. Any new member willing to join the group from the same Gung shall join so with realisation of the normal annual membership fees. Any members withdrawing from the membership shall not be eligible to claim any benefits and NWFPs resources collection from the Management area unless it is a genuine case (illness or famine). Member’s withdrawal on genuine ground shall be verified and if found genuine shall have right to claim 50% of his share from the group. During the time when new non-members want to join a short meeting shall be held for the necessary verification. 10.2 Responsibilities of members: Actively participate in general assembly to be held once in a year. Collect NWFPs and inform/hand over to committee for proper record and marketing. Support the chairperson and committee members to protect the resources from illegal activities. Ensure sustainable harvesting of NWFPs resources.		
11. Executive committee			
11.1. Chairperson	Name of Chairperson: Kinley Gyeltshen		
	11.1.1 Responsibilities of Chairperson Chair the meeting related to NWFPs management, utilization and marketing. Work in close coordination with other current prevailing Organizations, sectors and agencies, and seek assistance for the development of management group. To obtain permit for the sale of group’s/individual’s handicraft items and other NWFPs from CFO.		

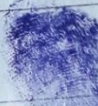
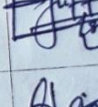
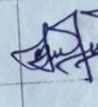
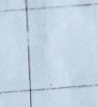
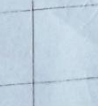
	Represent the group in meetings that will take place in other parts of the country and share new information and ideas acquired from the meeting with the group members on his return. Resolve disputes and conflicts arising among the group members and maintain good working relations. Coordinate the patrolling from time to time involving the members. Initiate annual planning with committee members and present to the group members during general assembly. Jointly sign with the Secretary and Treasurer on the cheque during the withdrawal of community fund.
11.2. Secretary	Name of Secretary: Joti Lal Gurung 11.2.1 Responsibilities of Secretary Officiate as chairperson. Inform the date, time and venue of meeting to all the group members. Keep minutes of NWFPs related meetings and make presentation at the end. Assist the chairperson in solving conflicts arising among members and maintain good working relations, and also in marketing of NWFPs. Keep proper record and data of NWFPs collected and marketed. Jointly sign with the Chairperson and Treasurer on the cheque during the withdrawal of community fund. Prepare monthly/quarterly or annual progress report and submit to the CFO.
11.3. Treasurer	Name of Treasurer: HarkaBdr Gurung 11.3.1 Responsibilities of Treasurer Keep proper records and accountings of NWFPs and assist the chairperson and secretary in marketing. Issue a receipt to the individual for quantity of NWFPs collected for sale. Keep annual financial transaction records and present to the group members during the general assembly. Collect and deposit the money in the NWFPs group account maintained with the nearest financial institution. Jointly sign with the Chairperson and Secretary on the cheque during the withdrawal of community fund. Prepare financial report and submit to the CFO as and when asked.

12. Compensation for executive Committee members	The committee members will serve for five years without any incentives. But if there are opportunities for trainings from the Government and other agencies, priority will be given to them as an incentive. It will be given on turn wise basis, if it is not work specific.
13. Membership Fee and purpose	Every member shall contribute annual membership fee of Nu. 100 (hundred) only. The amount will be deposited in the nearest bank and utilized the accumulated amount for the welfare of the NWFPs group members in the near future. The collect amount shall be used in case in any meeting or other activities of the group.
14. Community development fund and utilization of fund	The community development fund will be generated through deduction of certain percentage from the sale amount of NWFPs and handicraft items. For handicraft items, the deduction will be 10% of the total amount earned from each individual or as a whole after the marketing or sale of products annually. The fund shall be utilized for the welfare of NWFPs group members only. During emergencies (death), natural calamities etc, not more than 5% of amount will be given to the victim as “semso” upon discussion/meeting with the community. In case of utilization of the fund, a small discussion/meeting shall be held and the utilization of the fund shall be approved which shall then be used and later the bill/memo shall be presented to the group in the annual meeting.
15. Conflict Resolution	The chairperson and executive committee members shall resolve the conflict arising among the group members. If the conflict arises between executive committee members and the group members, the matter shall be forwarded to Gewog administration and Range office, which if unable to solve shall further be forwarded to the Prevailing Higher Authority(Court) If the problem arises between group members and other communities, the matter shall directly forward to Gewog Administration for settlement, and if it is not resolved at Gewog level, matter shall be further forwarded to the Dzongkhag Administration.

16. Penalties	Collection or harvesting of NWFPs for rural consumption purposes will remain free of royalty while for the sale of NWFP (Cane and Cane Shoots) including handicraft items shall require permit from CFO. Fines and penalties against the illegal activities within the Management area; NWFPs collection by him/her without the permission will be fined as per FNCRR, 2017, 2020 and 2021 and the products shall be seized. The fine should be paid off within 30(thirty days), in case, if failed, it shall be forwarded to Gewog Administration/Forest office/court or stopped him/her from availing the resources from the forest by the community through general assembly. Absent fees during meeting Nu.200 If a member of the NWFP Group violates the harvesting rules as set in this plan, he or she will not be allowed to harvest the NWFP anymore for the rest of the season and the following season. If there is any offence that is not mentioned above, it shall be dealt as per the latest rules and regulation of the Department of Forests and Park Services.
17. Authority of the Department	As per Chapter III, section 155 of FNCRR, 2017, the Department shall: Suo moto or on the request by the Non-Wood Forest Management Group (NWFMG) take action against violator of the Act, Rules or Management Plan; Inspect, monitor, review the records of and otherwise supervise the activity of the NWFP Management Group; and Take action including cancellation of the NWFP Management Group certificate if the NWFP-MG or any of its members who violate the provision of the Management Plan or any provision under this Chapter.

Annexure I: Membership list

Sl.no	Name	CID	Address	House no	Thram .no	Signature
1	phupzhemo	11315001220	Bana	GA-7-34	335	
2	Kenliy Tshering	10307000 343	Bana	GA-7- Neill	420	
3	Yeshey Zhemo	1030700 563	Bana	GA-7- 140	54-	
4	Ram Kumar Gurung	10307000410	"	G-7-51	79	
5	Kul Bdr Gurung	10307000524	"	Nil-58	403	
6	Chatur Nan Gurung	10307000900	"	G-7-47 47-440	490	
7	Harka Bdr Gurung	10307000460	"	G-7-57	117	
8	Migmar Lhamo	10307000387	"	G-7-38	54	
9	Bhijman Gurung	10307000403	"	G-7-50	77	
10	Passang Ngem	10307000311	"	GA-7-35	341	
11	Wangmo	10307000319	"	GA-7-36	52	
12	Passang Wangmo	10307000284	Bana	GA-7-77	672	
13	Santa Bir	10307000366	"			
14	Chagyal Wangmo	10307000587	"	Nil-13	19	
15	Kintey Dema	10307000343	"	H2:	68	

Sl.no	Name	CID	Address	House no	Thram .no	Signature
16	kesang Choden	10307000314	Bana	aliu-15		
17	Phub Dema	10307000283	11			
18	Gangchumo	10307000575	11	Mill-14	376	
19	Sungay Wangmo	10307000303				
20	Prem Bolr Ehung	10307000482	11	60	31	
21	Lok Bolr Ehung	10307000439	11	54	29	
22	Joli Lal	10307000479	11	59	30	
23	Jintey Ehung	10307000356	11	40	126	
24	Dal Bolr Ehung	10307000342 398	11	49	42	
25	Klon Bolr Ehung	10307000052	11	7	59	

Annexure II: Block description form for canes

Appendix 2: Block description form for canes

Date: 28-03-20

Village: 3a10

Block name: Block I

Name of cane area: Lalichok

Total area (ha): 36 Ha

Access	What is the walking distance from the nearest village to the block?		
	✓ Less than 3 hours	Within a day	More than a day
	What is the walking distance from the block to the nearest road?		
	✓ Less than 3 hours	Within a day	More than a day

Slope	Very steep	Steep	Moderate	Gentle
	>100%	70-100%	45-70% ✓	< 45 %

Forest description	Dominant and associate species, canopy cover:			
	① Dominant species are Castanopsis spp, beilschmiedia spp & Schima spp and Associate spp are Michelia spp and Exbucklandia spp. ② Canopy cover is more than 10%.			

Sub-block	No.	Density	Area (ha)	No of sample plots
	1			
	2			
	3			

Plantation/enrichment potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	1			
	2			
	3			

Current management	Current management practices in entire block:			
	✓ Clear harvesting, no sustainable harvesting method.			

Current threats	Threat	Description
	Fire	
	Grazing	
	Illegal exploitation	As locals harvest without approval.
	Over-exploitation	As locals don't have limit to harvesting.
	Other:	

Opportunities	Future opportunities for the entire block:			
	• Income generation and market development			

Appendix 2: Block description form for canes

Date: 25/3/20

Village: Salo, Bana

Name of cane area: Dorona

Block name: Block II

Total area (ha): 4.65 Ha

Access	What is the walking distance from the nearest village to the block?		
	☒ Less than 3 hours	☐ Within a day	☐ More than a day
	What is the walking distance from the block to the nearest road?		
	☒ Less than 3 hours	☐ Within a day	☐ More than a day

Slope	Very steep	Steep	Moderate	Gentle
	>100%	70-100%	45-70% ☒	< 45 %

Forest description	Dominant and associate species, canopy cover:			
	☒ Castanopsis & Balischemedia dominant. (Canopy cover more than 10%) ☒ Associate species are Michelia spp & Schima spp.			

Sub-block	No.	Density	Area (ha)	No of sample plots
	1		4.65	5
	2			
	3			

Plantation/enrichment potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	1	☒		
	2	☒		
	3	☐		

Current management	Current management practices in entire block:	
	☒ No technical technical and sustainable management ☒ conventional overharvesting method.	

Current threats	Threat	Description
	Fire	no threat from fire.
	Grazing	-
	Illegal exploitation	☒ Yes, it a threat as they harvest unsustainably for the sale in the near-market.
	Over-exploitation	☒ Un-sustainable harvesting of resources.
	Other:	

Opportunities	Future opportunities for the entire block:	
	☒ Income generation through shoots & development of various handicraft.	

Appendix 2: Block description form for canes

Date: 24-3-20

Village: Aetshipu

Block name: Block III

Name of cane area: Pakhaychok, Sukaypokai, &

Total area (ha): 69.6 Ha

Aetghilum.

Access	What is the walking distance from the nearest village to the block?		
	☒ Less than 3 hours	☐ Within a day	☐ More than a day
	What is the walking distance from the block to the nearest road?		
	☒ Less than 3 hours	☐ Within a day	☐ More than a day

Slope	Very steep	Steep	Moderate	Gentle
	>100%	70-100%	45-70%	< 45 %

Forest description	Dominant and associate species, canopy cover:			
	☒ Dominant species are castanopsis spp, Beilschmiedia spp and schima spp.			
	☒ Associate species of bamboos and Michelia spp too.			
	☒ Canopy cover more than 40%.			

Sub-block	No.	Density	Area (ha)	No of sample plots
	1		69.6	34
	2			
	3			

Plantation/ enrichment potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	1	☒		
	2	☒		
	3			

Current management	Current management practices in entire block:	
	☒ No specific management practise ☒ unsustainable management.	

Current threats	Threat	Description
	Fire	
	Grazing	
	Illegal exploitation	☒ As collected by people illegally for sale.
	Over-exploitation	☒ No control harvesting.
	Other:	

Opportunities	Future opportunities for the entire block:	
	☒ Introduction of nursery and re-introduction. ☒ Market product development and Income generation.	

Annexure III: Data Compilation Sheet

Data Compilation for Block I					
<i>PlectocomiaHimalayana</i> (Canes)					
Plot no	<2m	2-<10m	10-20m	L>20m(Cane<10)	L>20m(Cane>10)
1	10	0	0	0	0
2	0	0	0	0	0
3	12	0	2	0	5
4	8	4	3	5	3
5	4	0	0	2	0
6	16	2	2	0	2
7	17	2	1	3	4
8	14	2	1	3	5
9	18	4	2	4	3
10	8	2	3	7	5
11	0	0	0	0	4
12	0	0	0	0	0
13	11	4	0	2	6
14	7	1	2	2	4
15	9	0	4	2	4
16	0	0	0	0	0
17	0	0	0	0	0
18	7	1	1	0	4
19	0	0	0	0	0
20	0	0	0	0	0
21	6	0	0	6	8
22	0	0	0	0	0
23	0	0	0	0	0
24	0	0	0	0	0
25	0	0	0	0	0
26	1	0	0	3	3
27	0	0	0	0	0
28	0	0	0	0	0
29	0	0	0	0	0
30	0	0	0	0	0
31	0	0	0	0	0
32	6	1	0	2	4
33	0	0	0	0	0
34	0	0	0	0	0
	148	23	21	41	64

Data Compilation for Block I					
<i>Plectocomiahimalayana</i> (Shoots)					
Plot no	<2m	2-<10m	10-20m	L>20m(Cane<10)	L>20m(Cane>10)
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	2	3	0
4	0	4	2	4	0
5	0	0	0	0	0
6	0	3	1	0	0
7	0	4	2	3	0
8	0	2	1	4	0
9	0	5	2	3	0
10	0	3	5	3	0
11	0	0	3	2	0
12	0	0	0	0	0
13	0	8	1	13	0
14	0	5	2	0	0
15	0	4	2	2	0
16	0	0	0	0	0
17	0	0	0	0	0
18	0	0	3	1	0
19	0	0	0	0	0
20	0	0	0	0	0
21	0	6	4	0	0
22	0	0	0	0	0
23	0	0	0	0	0
24	0	0	0	0	0
25	0	0	0	0	0
26	0	1	0	0	0
27	0	0	0	0	0
28	0	0	0	0	0
29	0	0	0	0	0
30	0	0	0	0	0
31	0	0	0	0	0
32	0	5	0	10	0
33	0	0	0	0	0
34	0	0	0	0	0
Total	0	50	30	48	0

Data Compilation for Block II					
<i>Plectocomiahimalayana (Canes)</i>					
Plot no	<2m	2-<10m	10-20m	L>20m(Cane<10)	L>20m(Cane>10)
1	27	2	1	6	7
2	24	7	8	11	7
3	18	11	0	6	9
4	25	3	1	5	17
5	19	0	0	4	19
Total	113	23	10	32	59
Data Compilation for Block II					
<i>Plectocomiahimalayana (Shoots)</i>					
Plot no	<2m	2-<10m	10-20m	L>20m(Cane<10)	L>20m(Cane>10)
1	0	5	2	7	0
2	0	10	5	5	0
3	0	8	0	5	0
4	0	3	2	11	0
5	0	1	0	5	0
Total	0	27	9	33	0

Data Compilation for Block III					
<i>Plectocomiahimalayana (Canes)</i>					
Plot no	<2m	2-<10m	10-20m	L>20m(Cane<10)	L>20m(Cane>10)
1	3	0	0	0	1
2	0	0	0	0	0
3	0	0	0	0	0
4	24	0	7	1	16
5	22	0	0	4	14
6	10	3	0	0	9
7	26	3	0	10	15
8	25	0	0	4	14
9	17	0	8	4	17
10	0	0	0	0	0
11	11	0	0	13	4
12	16	2	3	23	11
13	7	2	0	4	24
14	26	0	1	0	13
15	2	0	0	2	12

16	2	1	0	1	2
17	12	2	0	3	18
18	8	0	0	2	12
19	2	1	0	1	0
20	0	0	0	0	0
21	5	2	2	3	6
22	1	0	0	0	4
23	20	0	0	9	23
24	16	0	0	4	23
25	6	0	2	2	5
26	10	0	2	7	9
27	11	2	1	6	7
28	16	2	0	5	12
29	17	0	9	5	8
30	10	2	5	2	1
31	9	0	0	0	0
32	4	0	3	0	3
33	5	0	0	0	2
34	4	2	1	3	2
35	7	0	0	0	15
36	6	1	2	5	6
37	14	1	0	10	20
38	5	0	1	1	2
39	13	0	5	2	10
40	4	0	0	10	4
41	0	0	0	0	0
42	9	0	0	0	10
43	10	0	0	4	8
44	10	0	0	0	7
45	0	0	0	0	5
46	19	0	2	1	10
47	3	4	8	0	12
48	5	0	0	2	11
49	10	0	6	0	5
50	7	1	3	0	6
51	13	0	2	8	17
52	8	2	2	0	10
53	1	2	0	0	5
54	14	0	0	0	19

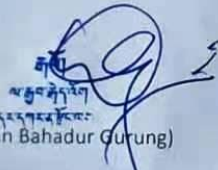
55	13	0	0	0	7
56	0	0	0	0	0
57	7	0	0	10	11
58	13	0	0	0	16
59	22	0	0	14	10
60	6	0	0	2	0
61	22	2	9	16	6
62	6	0	0	1	7
63	7	0	0	5	6
64	14	0	0	6	18
65	0	0	2	2	0
66	27	0	0	14	7
67	33	0	0	26	15
68	5	0	0	4	5
69	23	1	0	6	6
Total	703	38	86	261	583

Data Compilation for Block III					
<i>Plectocomiahimalayana (Shoots)</i>					
Plot no	<2m	2-<10m	10-20m	L>20m(Cane<10)	L>20m(Cane>10)
1	0	6	1	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	0	7	0	2	0
5	0	4	0	0	0
6	0	7	4	10	0
7	0	4	6	0	0
8	0	1	1	0	0
9	0	13	7	4	0
10	0	0	0	0	0
11	0	1	3	0	0
12	0	10	8	2	0
13	0	9	5	12	0
14	0	7	3	6	0
15	0	6	10	12	0
16	0	1	0	0	0
17	0	1	1	1	0
18	0	0	2	2	0

19	0	3	0	0	0
20	0	0	0	0	0
21	0	2	2	0	0
22	0	0	0	4	0
23	0	8	11	2	0
24	0	11	2	3	0
25	0	2	2	1	0
26	0	0	4	1	0
27	0	1	1	8	0
28	0	5	5	6	0
29	0	0	7	0	0
30	0	5	4	5	0
31	0	0	3	0	0
32	0	1	0	0	0
33	0	2	1	10	0
34	0	1	1	6	0
35	0	3	5	5	0
36	0	1	4	0	0
37	0	5	5	9	0
38	0	1	5	5	0
39	0	4	3	6	0
40	0	0	0	10	4
41	0	0	0	0	0
42	0	10	3	2	0
43	0	10	8	0	0
44	0	4	2	4	0
45	0	1	0	1	0
46	0	2	3	1	0
47	0	4	5	2	0
48	0	2	1	5	0
49	0	3	8	3	0
50	0	1	3	6	0
51	0	2	14	6	0
52	0	2	5	10	0
53	0	2	0	0	0
54	0	8	2	12	0
55	0	0	2	2	0
56	0	0	0	0	0
57	0	2	1	2	0

58	0	14	22	0	0
59	0	17	15	4	0
60	0	6	9	0	0
61	0	6	6	4	0
62	0	1	3	1	0
63	0	8	2	0	0
64	0	16	13	4	0
65	0	3	0	0	0
66	0	9	6	2	0
67	0	17	25	2	0
68	0	4	0	2	0
69	0	9	4	6	0
Total	0	295	278	213	0

Annexure IV: Application for the establishment of NWFP group from Gewog Administration

	དཔལ་ལྷན་འབྲུག་གཞུང་། ཆེད་འོག་བདག་སྐྱོང་། ROYAL GOVERNMENT OF BHUTAN GEWOG ADMINISTRATION LAJAB GEWOG
Ref.No. LJG/F(05)/2021/ 349	Date:22/06/2021
To, Range Officer, Drujeygang,Dagana	
Sub: <u>Formation of Cane Shoot Group.</u>	
Dear Sir,	
Lajab Gewog Administration would like to request for an approval to form a Cane Shoot Group at Bana Chiwog as they have got available cane at Bana Community forest. They will sale the cane shoot under sustainable terms and condition that they will build the bylaws during the formation of Group.	
Therefore, we would be grateful to your consideration and necessary action please.	
Thanking You	
Yours sincerely,	
 (Dhan Bahadur Gurung)	
Gup : Lajab Gewog	
Gup (17762445/77226444) Mangmi (77309655) Gewog Administrative Officer (77732132)	

